

Stabilizing a topological radius exposes an action coefficient

A quadratic/quartic field energy has a finite preferred scale within an explicit degree-one family. Its energy-times-radius divided by wave speed is proportional to the quartic coefficient divided by that speed. Scaling both field coefficients leaves the classical dynamics unchanged while scaling this action. Size stabilization succeeds; absolute action normalization remains an independent input.

Status: Q05 exploratory derivation, not an accepted-ledger claim. Independent review and a bounded librarian comparison precede any promotion. The scale minimum below is a trial-family result, not a proof of a full field minimizer.

1. Field, sector and kinetic term

Let $n : \mathbb{R}^3 \rightarrow S^3 \subset \mathbb{R}^4$ be smooth, approach a fixed point at spatial infinity, and have finite energy

$$E[n] = \int \left[a \sum_i |\partial_i n|^2 + b \sum_{i < j} \left(|\partial_i n|^2 |\partial_j n|^2 - (\partial_i n \cdot \partial_j n)^2 \right) \right] d^3 x. \quad (1)$$

Here $a > 0$ has units energy/length, $b > 0$ has units energy times length, and $c > 0$ has speed units. Compactifying the domain defines the integer degree N . This is the massless Skyrme-type quadratic/quartic construction written in real sphere coordinates; its standard ingredients have prior art.

To specify physical time, choose the conservative Lagrangian

$$L = \int \left[\frac{a}{c^2} |\dot{n}|^2 + \frac{b}{c^2} \sum_i \left(|\dot{n}|^2 |\partial_i n|^2 - (\dot{n} \cdot \partial_i n)^2 \right) \right] d^3 x - E[n]. \quad (2)$$

The velocity quadratic form is positive. This is the usual Lorentz-invariant quadratic/quartic completion, with coordinate $x^0 = ct$; all equations here use physical time t . We require smooth solutions where they exist. This specification supplies neither global nonlinear well-posedness nor a hard speed theorem for arbitrary backgrounds.

For completeness, if $\lambda_i \geq 0$ are the singular values of dn , the static density is $a \sum_i \lambda_i^2 + b \sum_{i < j} \lambda_i^2 \lambda_j^2$. Pairing $a \lambda_i^2$ with $b \lambda_j^2 \lambda_k^2$ and applying the arithmetic-geometric mean inequality gives density at least $6\sqrt{ab} \lambda_1 \lambda_2 \lambda_3$. Since the unit three-sphere has volume $2\pi^2$, integration yields the standard degree estimate

$$E \geq 12\pi^2 \sqrt{ab} |N|. \quad (3)$$

The absolute Jacobian integral bounds the absolute degree even when local orientations vary. A positive sector energy is distinct from a vacuum excitation gap or an action bound.

2. An explicit size test

Write $y = x/R$, $r = |y|$, and use the inverse stereographic profile

$$n_R(x) = \frac{(2y_1, 2y_2, 2y_3, r^2 - 1)}{1 + r^2}, \quad R > 0.$$

It has $|N| = 1$, and R is the sphere's equator radius in the domain. All three dimensionless singular values are $2/(1 + r^2)$. Thus

$$A = 48\pi \int_0^\infty \frac{r^2 dr}{(1 + r^2)^2} = 12\pi^2, \quad B = 192\pi \int_0^\infty \frac{r^2 dr}{(1 + r^2)^4} = 6\pi^2.$$

The integrals follow directly from $r = \tan u$, $0 < u < \pi/2$. Spatial scaling gives

$$E(R) = aAR + \frac{bB}{R}, \quad R_* = \sqrt{\frac{bB}{aA}} = \sqrt{\frac{b}{2a}}, \quad E(R_*) = 12\sqrt{2}\pi^2\sqrt{ab}. \quad (4)$$

Unlike Q04, shrinking this family sends its energy to infinity. Expansion also costs energy. This proves a unique minimum along its scale direction, with positive second derivative $2bB/R^3$. General shape perturbations remain outside this one-parameter test. In particular, n_{R_*} is a trial field, not asserted to solve the static Euler–Lagrange equation.

3. Where the action enters

Define the same type of diagnostic as Q04: evaluate the magnitude of static Lagrangian action over the profile’s crossing duration $T_R = R/c$,

$$\mathcal{A}_R = E(R)R/c = (aAR^2 + bB)/c. \quad (5)$$

For off-shell static trial fields this is a functional diagnostic, not an on-shell trajectory action. It has infimum $6\pi^2b/c$ within this family; at its energy-optimal scale it equals

$$\mathcal{A}_{R_*} = 12\pi^2b/c. \quad (6)$$

The product b/c already has action units. More generally, setting $x = \ell y$, $t = \ell s/c$ with $\ell = \sqrt{b/a}$ writes the full action as

$$S = \frac{b}{c} \tilde{S}[n], \quad (7)$$

where the dimensionless functional has unit quadratic and quartic coefficients. This includes the time-derivative terms, not just static energy. It identifies exactly which normalization a proposed quantum phase would use.

There is also an exact counter-test. Replace (a, b) by $(\eta a, \eta b)$, $\eta > 0$. The length ℓ , field equations, degree, and characteristic vacuum speed remain unchanged. Energies and actions multiply by η . Every positive η defines a nondegenerate model, and actions tend to zero as $\eta \downarrow 0$. No claim about dynamics at the degenerate endpoint is needed. This varies the physical stiffnesses; it does not deny fixed-coefficient sector bounds. It shows that field equations and topology alone do not fix their common mechanical normalization.

4. Physical spectrum and next decision

Near a constant vacuum, write $n = (\pi, \sqrt{1 - |\pi|^2})$. To quadratic order,

$$L^{(2)} = a \int [c^{-2}|\dot{\pi}|^2 - |\nabla\pi|^2] d^3x, \quad \Omega(k) = c|k|.$$

The quartic term has no quadratic vacuum contribution. On infinite space there are arbitrarily low frequencies. Fixing a nonzero topological sector and stabilizing its radius therefore leaves this vacuum spectrum gapless. It also supplies no mechanism forcing a trajectory to carry nonzero degree.

Retain gradient competition as a conservative size-exclusion mechanism. Its action coefficient is explicit, while the vacuum-gap question remains separate. After two topology steps, park further adjustable stabilizers. The next physical test should couple sectors that initially have independent normalizations: determine whether reciprocal energy exchange can constrain their ratio and whether one overall action normalization survives. This can turn the common-normalization obstruction into a concrete composition test, rather than adding another term whose coefficient contains the answer.

Source boundary

The discovery lead is *Topological energy bounds in generalized Skyrme models*, arXiv:1311.2939, publisher abstract. Coverage here is the publisher/search abstract only: it identifies opposite Derrick scaling and topological bounds as established ingredients. Full-text formula matching and independent review are pending, not silently imported. The normalization, trial integrals, kinetic expansion and action comparison above are written calculations. No computational verification scripts were used.